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Dynamic Dempster Multi-Layer Perceptron for the prediction of admission patient in emergency department

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***Abstract*— The early identification of the patients' hospitalization at triage level within the Emergency Department (ED) presents a potential solution to reduce the risk of overcrowding and improve the quality of care. Thus, predicting patient out-come on arrival assists medical staff in the make of the appropriate patient pathway decision and so reduces the risk of medical error and complication of the patient's condition. Previous works don't consider the uncertainty of medical data while the management of this uncertainty is one of the most important and crucial tasks of medical information systems. Thus, we present in this paper an improved version of the classical prediction model by taking into account the uncertainty and by managing properly the missing information. In this context, we propose a new approach based on Dempster-Shafer theory and Dynamic Multi-Layer Perceptron algorithm. Our proposed approach deploys the correspondent neural network as follows: 1) computes for each input parameter the Basic Belief Assignment (BBA) that provides an assessment of the uncertainty pattern using the Dempster's rule; 2) deduces the correspondent weights based on the computed BBA, and 3) uses an appropriate transfer function to activate the next layer neurons. In this paper, we demonstrate the effectiveness of our proposed method by using a real ED database. We prove that our proposed approach manages efficiently the uncertainty of the medical data sources and missing information, so improves the decision making and reduces errors and complexity.**

Keywords*— *Medical prediction model, Dempster-Shafer Theory, Machine Learning, Neural networks, Uncertainty.

I. INTRODUCTION

The overcrowding poses a significant international problem in Emergency Departments (EDs) [1]. Some of the main reasons, are increased patient's demand, misdirected patient in the care process and a lack of hospital inpatient units [2]. Crowding may have a negative impact on patients at various stages of their ED journey as increase wait time, risk of mortality [3], and hazard of errors medical decision. The early prediction of the patient required hospitalization at triage can take measures to address such problem. This urges the hospitals to better manage the patients flow as well as the capacity and the demand for hospital beds [4]. Several medical data-based prediction models have been developed to classify patients at high risk of hospitalization. The goal is to avoid bottlenecks by better managing the attribution of hospital beds. The long patients waiting for beds downstream from the EDs creates overcrowding and increase the risk of mortality. Among these works, we find several techniques that are used classification algorithms [5], based on data as Logistic Regression (LR), neural networks and decision tree. The Standard Multi-Layer Perceptron (SMLP) presents the subject of extensive theoretical research for supervised classification. However, the disadvantage of SMLP in a real application of the Back-Propagation (BP) algorithm is the slow learning speed and the very high complexity. In recent works, a Dynamic Multi-Layer Perceptron (DMLP) algorithm [6] was created to improve the SMLP algorithms. The dynamism of many of the input neurons can reduce the complexity of this technique. For that, we will be inspired by this principle and we will deal with the problem of uncertainty. In addition, BP phase takes hundreds or thousands of times to get the weights from the network because it iteratively updates the weights between neurons. Moreover, the lack of medical information, inherent data sources and the incompleteness of medical knowledge causes of data uncertainty in the medical field. These problems always remain major limits for these models. The Dempster–Shafer theory of evidence (D-S theory, also known as evidence theory) is identified as an efficient model to measure the uncertainty, fuse information and manage decision-making problems in intelligent systems [7, 8, 9]. Dempster's combination rule is the most crucial instrument of D-S theory and it possesses several interesting mathematical properties. The uncertainty can describe the quality or clarity of the evidences. In this context, we propose an approach based on DMLP network and D-S theory called Dynamic Dempster Multi-Layer Perceptron (DDMLP). This approach is composed of two phases. The first phase corresponds to a forward algorithm including a new method of weight calculations based on uncertainty by D-S rules. The second phase corresponds to the application of the BP algorithm in the backward phase. The latter is a necessary phase to find its optimal parameters and reduce the error rate in this algorithm. In fact, the aim of this paper is (1) to propose a new approach of supervised classification by modelling of dynamic neural networks based on the D-S theory and considering data uncertainty while reducing the complexity to predict patient hospitalization at EDs triage, and (2) to evaluate the performance of proposed model and compare it with other classification algorithms. We conduct this work with the ED of Lille University Hospital Center (LUHC) in France in the context of a national project called Inter and Intra Hospital Logistics Optimization[1] supported by the National Research Agency (2019-2022).

II. RELATED WORK

First, Many previous studies were developed to predict hospital admission at the time of emergency triage using machine learning techniques. Most models include only information collected at time of arrival such as demographics data, vital signs, reason for arrival, nursing notes and early diagnosis [10], while some models include other additional features such as symptoms and Medical History (MH). In the literature, Graham et al. [5] developed three data mining methods to set up a prediction model using historical data. This latter improved the decision quality and reduced the errors rate but unstructured and uncertainty inputs data posed problems in

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classification techniques. Wang et al. [11] used machine learning techniques to predict ED admissions, they compared the capacity of Fuzzy Min-Max neural networks (FMM) to other standard data mining algorithms as the supervised classification with decision and regression trees (CART), SMLP, random forest and AdaBoost. Overall, the SMLP and Random Forest models were the most accurate with 80% accuracy and the FMM (with a genetic algorithm) with 77.97% accuracy but in spite of these efficient results, the risk of false positive data is still important due to inconsistent and missing medical data. In literature, previous studies have been limited for several reasons. To address the first challenge, several works are focused only on structured data and Text data collected [13]. Structured data such as the age, the gender, or arrival mode [3] and unstructured text data, such as MH. This latter is generally excluded due for filtering and operating difficulties, except some studies [12,20]. Zhang et.al [13] has used the Natural Language Processing to analyze, understand, and extract information from unstructured and semi structured data before classification step but the used words model ignored the order of the words and the specific contextual information. As for the second challenge, the same of data source [14] are inherent to the type of analysis performed. Also, the missing values in the datasets affect the predictive performance of the models [13] and the survey data may have some misclassification and ascertainment bias with ruling out the missing information [15]. Therefore, missed values, inherent and incomplete increase the uncertainty aspect that decreases the reliability and performance of classification results. In fact, the classification based on neural networks generally gives good results. It can be described as massively parallel computer systems composed of an extremely high number of neurons with numerous interconnections [16] and has been widely used in supervised classification and information retrieval [6]. The advantage of using a neural network is the ability to learn input-output relationships with complex non-linear variables. The SMLP guarantee the full connectivity between nodes of a previous layer and nodes of the next layer and take the BP algorithm [16] as the main learning algorithm. Weights are used to establish the connection between the nodes of a previous layer and those of a next layer. However, despite many advances, the disadvantage of SMLP in a real application of the BP algorithm is the very high computational complexity. However, the DMLP algorithm [6] is created to improve the learning speed and to reduce this complexity by eliminating missing values in the calculations. It is an efficient method but the principal problem is the initialization of weights that is always random which brings the BP algorithm to perform several iterations to reach an adequate optimization. Hence, we propose in this paper to calculate in an accurate way the weights of neurons using the D-S theory that computes the evidences of each input data in order to drive more efficiently the learning process.

III. THE PROPOSED DDMLP METHOD

In this section, we expose the D-S theory before presenting our DDMLP approach for the prediction of the patient hospitalization.

A. The Dempster Sharfer theory (D-S theory)

The theory of belief functions is derived from Dempster's work on the lower and upper bounds of a family of probability distributions [7]. It was developed by Shafer [8] who showed the interest of belief functions as a formalism of representation of uncertainty. The principle of belief theory is based on the manipulation of mass functions on subsets and not on singletons as in probability theory. These functions usually have values in [0,1], and are called mass functions or elementary masses or belief masses. Let us consider the set Ω called the frame of discernment with $\Omega = \{x_1, x_2, \ldots, x_n\}$. It corresponds to all the possible decisions or answers to a given question as the set of all possible variables in a classification problem. An element of evidence is represented by a mass function m defined from the powerset of Ω to [0,1]. This powerset is commonly denoted by $P(\Omega)$ or 2^{Ω}, corresponding to the set of all possible subsets of Ω, including the empty set. Thus, it is necessary to add all the possible intersections between two elements in 2^{Ω}. An element x_i of Ω must respect its exclusivity; In D-S theory [8], a basic belief assignment (BBA), corresponding to a source of information, is a mass function with:

$$\sum_{A \in 2^{\Omega}} m(A) = 1 \quad \text{And } m(\emptyset) = 0 \tag{1}$$

The m(A) is interpreted as a measure of belief allocated exactly to the X ∈ A hypothesis, and to no more specific hypothesis. It is called BBA or BPA (basic probability assignment) or BOE (body of evidence). If m(A) > 0 then A is called a focal element, and the set of all focal elements is named a Body Of Evidence (BOE). For two available independent sources m_p and m_q, we can use the Dempster's combination rule by the combined evidence as follows (2):

$$m_{p,q} = m_p \oplus m_q$$
$$m_{p,q}(A) = (1-k)^{-1} \sum_{B \cap C = A} m_p(B) m_q(C) \quad \text{And if } A \neq \emptyset \quad m(\emptyset) = 0 \tag{2}$$

Where A, B and C are elements in 2^{Ω} and k corresponds to the conflict degree between the mass sets m_p and m_q with:

$$\kappa = \sum_{B \cap C = \emptyset} m_p(B) m_q(C) \tag{3}$$

The normalization factor (*1-k*) allows to ignore conflicts. So, it is not appropriate in the case of high conflict between different sources that give either contradictory information or give the same information with incorrect semantics. In this context, when the information derived from real data related to a patient state of health, this information should not be conflicting. In D-S theory, the belief function of A denoted Bel(A) is defined as:

$$Bel(A) = \sum_{B \subseteq A \neq \emptyset} m(B) \tag{4}$$

The plausibility function of A, denoted Pl(A) is defined as follow:

$$Pl(A) = \sum_{B \cap A \neq \emptyset} m(B) \tag{5}$$

Where $\forall A \in 2^{\Omega}$, Bel(A) represents the total belief degree in A, justified by the considered element of evidence. It corresponds to the lower bound of the belief degree susceptible to be allocated to A. Pl(A) corresponds to the upper bound of the belief degree susceptible to be allocated to A, considering new information. So, the probability of A, noted by P(A) is in [Bel(A), Pl(A)]. D-S theory is a technique that calculates the uncertainty related to a source or a combination of sources of information. A very high uncertainty proves a total ignorance of the given information. As explained above, we suppose in this paper that there is not significant conflict between sources of information because D-S theory is not effective in this case [17]. In the following part, we detail our proposed method.

B. Dynamic Multi-Layer Perceptron based on D-S theory (DDMLP)

In this part, we introduce our DDMLP algorithm with the classification pattern enforced. As explained previously, we choose to use the SMLP algorithm which usually requires a static number of input neurons. Generally, the network is expected to process a set of input vectors with a chosen maximum number of input parameters. However, the additional calculations required for unused inputs increase the processing time and decrease the speed learning. In this context, Mustapha et.al [6] proposed to ignore the unused input neurons to reduce the input vector size. In this way, these unused neurons that present missing data don't affect the calculations within the network. Inspired of from this idea, we propose the DDMLP which corresponds to a neural network composed of an input layer with a flexible neuron number and a hidden layer with neurons calculated according to the number of input neurons. If we consider that $N_{i,t}$ as the number of neurons of the layer t of the patient P_i, our network has two dynamic layers with $N_{i,1}$ input neurons corresponding to the number of non-empty medical parameters for the patient P_i and $N_{i,2}$ hidden neurons calculated by the Equation (6). We propose this formula because we merge each two successive parameters in the hidden layer to calculate the weights. This proposal

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is a hypothesis that we have proposed as a method for combining data sources and then in future work, we can improve it by studying the correlations between the data variables. Then, the data fusion allows us to find the mass of information source for a relation between two variables. In this way, we use only active neurons containing data in the calculation of the weights. For each patient Pi, an instance of DDMLP is created. Each instance is composed of an input layer, a hidden layer and an output layer. The number of neurons of the output layer is noted by Ns with Ns =1 since the output layer contains a single neuron.

$$N_{t,2} = |\frac{N_{t,1}+1}{2}| \quad (6)$$

Our model builds neurons based on the valid data for each patient case Pi. However, compared to Mustapha et.al [6], hidden layer neurons are dynamic in our neural network. We show in the next section how to compute the weights of the neurons, considering the sources of information in accordance with their relevance and so the uncertainty of data. Weights are called Weight Belief Assignment (WBA), by analogy with the BBA in D-S Theory.

1) Weight Belief Assignment (WBA)

The main idea of this work is to consider the data uncertainty in the processing and learning part. In this context, the weight (represented by *WBA*) corresponds to the amount of information gain that continues to propagate until the overall objective is reached. The figure 1 corresponds to a graphic representation of our proposed approach.

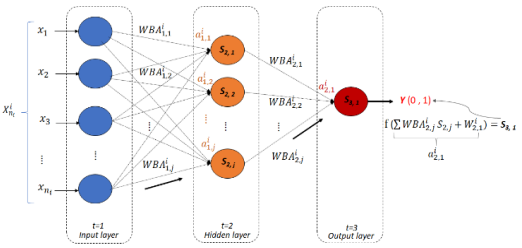


Fig. 1. DDMLP forward phase illustration.

We note by $WBA^i_{t,j}$ the weight associated to the patient P_i in the layer t according to the source (or combination of sources) j. In this way, weights of the network are transformed according to the uncertainty and then propagated through the network by the Dempster combination rule. The goal of our DDMLP is to forecast the patient outcome according to the available correspondent data. Since our frame of discernment Ω set corresponds to all the orientation possibilities d_z of one patient after the ED care where $z \in \{1,2\}$, we take an instance of these possibilities $\Omega = \{d_1, d_2\}$ with d_1 corresponds to the discharge of the patient and d_2 to his hospital admission. These decisions forecasting are very important in reducing medical errors, saving patients' lives and managing ED overcrowding. Each medical parameter (such as symptoms and MH) is considered as information source j to calculate the information mass for each patient case P_i. These data help to formulate the answers to our ED problem by managing the uncertainty of each given variable. $a^i_{t,j}$ presents the output of the neurons of the layer t, allowing the calculation of the inputs of the next the layer $t+1$ and $S_{t,j}$ defines the input variables of the neurons in the following layer t. The result y of our approach is a Boolean value, either 0 (d_1) or 1 (d_2). The decision to discharge patients is very complex because his health condition can suddenly deteriorate, which can cause sudden and premature death. Hence the importance to propose an intelligent decision support system at the triage level.

a) WBA in input layer

As explained above, $WBA^i_{t,j}$ corresponds to the weight associated to the patient P_i in the layer t according to the source (or combination of sources) j. So, the weight $WBA^i_{1,j}$ in the input layer is defined in our network by the Equation (7).

$$WBA^i_{1,j} = \frac{m^i_j(d_2) - m^i_j(d_1)}{m^i_j(\Omega)} \quad (7)$$

Where: m^i_j corresponds to the mass function (1) related to the information source j (medical parameter) of the patient P_i (medical case). So, $m^i_j(d_z)$ measures the evidence of the decision d_z with the information source j for the patient P_i ($z \in \{1,2\}$). This weight is all the more important as the chance of being hospitalized for the patient P_i is important according to the source j. Moreover, the greater the overall uncertainty, the lower this weight. The uncertainty of the information source j concerning the patient P_i in regard to the decision d_z is obtained with the Equation (8).

$$m^i_j(\Omega) = \prod_z m^i_{j,z}(\Omega)$$
$$\text{With } m^i_{j,z}(\Omega) = 1 - m^i_j(d_z) \quad z \in \{1,2\} \quad (8)$$

b) WBA in hidden layer

We define the weights $WBA^i_{2,j}$ in hidden layer on our network by the Equation (9). The only element that changes here is the source.

$$\mathrm{WBA}^{\mathrm{i}}_{2,\mathrm{j}} = \frac{m^i_j(d_2) - m^i_j(d_1)}{m^i_j(\Omega)} \quad (9)$$

Where j = j'⊕ j" and $m^i_{j',j''} = m^i_{j'} \oplus m^i_{j''}$ according to the Dempster combination rule (2). Contrary to the calculation of the weight of the previous Equation (7), we are not dealing with a single source but to the combination of two sources of information j' and j'' (randomly or heuristically selected). So, here j represents the combination of a couple of sources j' and j''. We apply the same rules to calculate the hidden layer weights but, in this step, we consider the combined evidence of the union of two information sources instead of a single source because this combination can give a semantic relationship between the sources allowing to reduce the uncertainty. For example, a 70 years old man has cardiological symptoms, he has a high possibility of being hospitalized.

2) The proposed DDMLP Algorithm

We propose in this paper the DDMLP algorithm (see Algorithm 1) which calculates each neuron weight using the Equations (7) and (9) then uses the usual Sigmoid activation function to get the layer outputs because the classification in this model is binary. The table 1 summarizes the used variables of our proposed DDMLP algorithm.

TABLE I. USED VARIABLES IN DDMLP ALGORITHM

Variable	Signification
x^i_j	Medical data j of the patient P_i
$X^i_{n_i}$	Vector of x^i_j with 1<j<n_i with n_i the number of medical data of each patient P_i.
T	Index of a layer (t∈{1,2,3})
$N_{i,t}$	Number of neurons of the layer t for the patient P_i
$a^i_{t,j}$	Output of the layer t corresponding to the source (or combination of sources) j of the patient P_i.
A_t	Intermediate output variable to compute $a^i_{t,j}$.
J	The information source of medical data such as age and blood pressure.
$WBA^i_{t,j}$	Weight associated to the patient P_i in the layer t according to the source (or combination of sources) j.

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$S_{t,j}$	Sigmoid variable corresponding to the layer t and information source (or combination of sources) j
$m_j^i(d_z)$	The evidence of the decision d_z with the information source (or combination of sources) j for the patient P_i
$W_{t,j}^i$	Dynamic DDMLP bias in the layer t according to the source (or combination of sources) j of the patient P_i.

Algorithm 1 DDMLP algorithm (forward Pass)

Input: $X_{n_i}^i$ \\ *Input variable vector*

Output: Boolean \\ *(0 = patient discharged) (1 = patient hospitalized)*

1: $N_{i,t} \leftarrow 0$, $A_t \leftarrow 0$, $S_t \leftarrow 0$, $\forall t \in \{1,2,3\}$

2: **For** each x_j^i of X_n^i **do**

3: **If** x_j^i is not null **then** \\ *Calculate the number of neurons in input layer*

4: $N_{i,1} \leftarrow N_{i,1} + 1$

5: **end If**

6: **end For** \\ *Calculate the number of neurons in hidden layer*

7: $N_{i,2} \leftarrow (N_{i,1} + 1)/2$

8: **For** t=1 to 3 **do**

9: **If** t=1 **then** \\ *In input layer*

10: **For** j=1 to $N_{i,t}$ **do** \\ *Calculate the weight of each neuron in the input layer*

11: $WBA_{t,j}^i \leftarrow \frac{m_j^i(d_2) - m_j^i(d_1)}{m_j^i(\Omega)}$ \\ *Calculate the vector weight*

12: $A_t \leftarrow A_t + x_j^i * WBA_{t,j}^i$

13: **end For**

14: **end If**

15: **If** t=2 **then** \\ *In hidden layer*

16: **For** j=1 to $N_{i,t}$ **do**

17: $W_{t-1,j}^i \leftarrow$ random \\ *The dynamic biais*

18: $a_{t-1,j}^i \leftarrow A_{t-1} + W_{t-1,j}^i$ \\ *Compute the outputs of the input layer*

19: $S_{t,j} \leftarrow \frac{1}{1+e^{-a_{t-1,j}^i}}$ \\ *Calculate the inputs of the hidden layer*

20: $WBA_{t,j}^i \leftarrow \frac{m_j^i(d_2) - m_j^i(d_1)}{m_j^i(\Omega)}$ \\ *Compute the weight of each neuron in the hidden layer*

21: $A_t \leftarrow A_t + S_{t,j} * WBA_{t,j}^i$

22: **end For**

23: **end If**

24: j$\leftarrow$1

25: **If** t=3 **then** \\ *In the output layer*

26: $W_{t-1,j}^i \leftarrow$ random

27: $a_{t-1,j}^i \leftarrow A_{t-1} + W_{t-1,j}^i$ \\ *Calculate the outputs of the hidden layer*

28: $S_{t,j} \leftarrow \frac{1}{1+e^{-a_{t-1,j}^i}}$ \\ *Compute the result of the output neuron*

29: **end If**

30: **end For**

31: **If** $S_{t,j}$ <=0,5 **Then** \\ *The decision d_1 or d_2*

32: return 0 \\ d_1

33: **Else**

34: return 1 \\ d_2

35: **End If**

This DDMLP algorithm retrieves as input the vector of the n_i medical data of each patient P_i and as explained previously, the weights are calculated on the basis of two parts: the uncertainty of the data sources and the measurement of the proof of this data used according to the decision d_z ($z \in \{1,2\}$). These computed weights are crucial for the calculation of the neurons inputs/outputs and also for keeping the relationship and the propagation of information between the layers. This algorithm starts by calculating the number of neurons of the input layer and of the hidden layer, the last (output) layer contains only one neuron. Then layer by layer (starting with the first layer), we calculate the weights and the outputs of each neuron, and finally the inputs of the next layer using the sigmoid function. For the output's computation, we propose to initialize randomly the variables $W_{t,j}^i$ which is equivalent to the fixed bias of the SMLP algorithm. The purpose here is to counterbalance the non-random assignment of weights for the learning process. So, $W_{t,j}^i$ is dynamic and plays a very important role in the backward phase to correct the error, so that we can keep their updates in the next phase. The backward pass starts by comparing the result of the forward algorithm y with the real result y', here it applies the BP algorithm to measure the error and modify the bias $W_{t,j}^i$ until optimizing the algorithm parameters (Fig. 2).

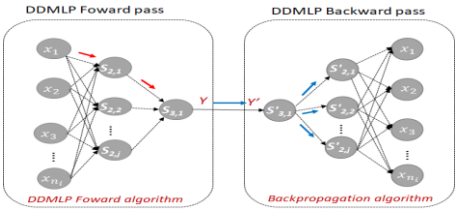


Fig. 2. DDMLP approach illustration.

After choosing randomly the bias $W_{t,j}^i$ (between -0.1 and 0.1) of the network, the backpropagation algorithm is used to compute the necessary corrections. The BP algorithm can be decomposed in the following four steps:

1) Feed-forward computation.
2) Backpropagation to the output layer
3) Backpropagation to the hidden layer
4) Bias updates.

This algorithm is stopped when the value of the error function E (10) has become sufficiently small.

$$E = \frac{1}{2} \sum_{i=1}^{\rho} |y'_i - y_i|^2 \quad (10)$$

Where y_i is of the current output set for each input data x_i presented to this network, the latter produces a result y'_i generally different from the target y_i. For $i = 1 \ldots p$ using a learning algorithm (BP), the goal is to repeat the correction process until Y' tends towards Y.

IV. EXPERIMENTATION

The present work is proposed in the context of the project OIILH (2019-2020): Optimization for Inter and Intra Logistics in Hospital. This project is supported by the French research agency and referenced by ANR-18-CE19-0019. A real ED database from the LUHC in France is used for our experimental study. It is a 4 years' database including the triage and orientation data for 2016 and 2020. This database we currently hold contains all the data relevant to the decision. We filtered then normalized this data in such a way to facilitate the usage of our proposed method on homogeneous data (e.g. transformation textual data into binary data). To apply the proposed method, we defined the decision variables for ED patient admission, the frame of discernment Ω is the set of decisions (d_z) for each ED patient P_i where ($z \in \{1,2\}$): d_1 is the decision of return at home and d_2 is the hospitalization. For the dataset, we used 225,392 cases of each class d_z for training and 111,064 patients' cases for testing. Our database contains 59% patients who returned home and 41% inpatients. The information sources present the input medical data: age, Patient Status (PS), Arrival Mode (AM), ESI (Emergency Severity Index), temperature, Chief Complaint (CC), MH, etc. In what follows, we firstly explain the deployment of the proposed method on a small example before demonstrating the its advantage by comparing it with other existent methods in a second step.

A. Simulation example: the ED problem case

In this section, we show the execution of our DDMLP algorithm step by step with a small size problem contains 5 patients. The table

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2 shows the BBA of each medical input x_j^1 of the patient P_1 respecting the equation (1).

TABLE II. BBA'S OF MEDICAL DATA

Data j	$m_j^1(d_1)$	$m_j^1(d_2)$	$m_j^1(\Omega)$
1: AM = SAMU	0.21	0.44	0.35
2: PS = Valid	0.35	0.39	0.26
3: age = 70	0.33	0.48	0.19
4: MH = cardio	0.31	0.35	0.34
5: ESI = 2	0.18	0.59	0.23
6: CC = malaise	0.29	0.41	0.30

Next, the algorithm calculates the WBA of each patient P_i medical parameter with the Equation (7) (table 3).

TABLE III. WBA CALCULATION OF THE INPUT LAYER

$WBA_{1,j}^i$	P_1	P_2	P_3	P_4	P_5
$WBA_{1,1}^i$	0.657	0.266	0.144	0.122	0.238
$WBA_{1,2}^i$	0.153	0.544	0.376	0.355	-0.123
$WBA_{1,3}^i$	0.789	-0.78	0.576	0.78	0.345
$WBA_{1,4}^i$	0.117	-0.01	-0.876	-0.87	0.45
$WBA_{1,5}^i$	1.782	0.122	0.456	0.233	-0.456
$WBA_{1,6}^i$	0.4	0.144	-1.33	0.877	1.12

In this stage, the Equation (2) is used to compute the combined mass of each couple of sources (table 4). The latter shows each combination between two successive sources in table 5 (e.g. j = 1 corresponds to the combination of masses between AM and PS sources). Then, we calculate the activation function using the outputs of the previous layer $a_{1,j}^i$.

TABLE IV. INPUT CALCULATION OF THE HIDDEN LAYER

Data j	$S_{2,j}$	$m_j^1(d_1)$	$m_j^1(d_2)$	$m_j^1(\Omega)$
1	-0.839	0.018	0.148	0.09
2	-0.795	0.081	0.040	0.059
3	-0.745	0.142	0.02	0.05

Now, the WBA is computed in the hidden layer using Equation (9) (table 5).

TABLE V. WBA CALCULATION OF THE HIDDEN LAYER.

$WBA_{2,j}^i$	P_1	P_2	P_3	P_4	P_5
$WBA_{2,1}^i$	1.44	0.79	0.03	-2.44	1.74
$WBA_{2,2}^i$	-0.694	-1.33	-2.07	0.22	0.33
$WBA_{2,3}^i$	-2.44	0.56	0.66	1.42	-0.66

Then, we calculate the outputs $a_{2,j}^i$ according to the weights $WBA_{2,j}^i$. Then, we randomly select $W_{3,1}^1$= 0.1 to compute the input of the output layer $a_{2,1}^1$ (table 6).

TABLE VI. CALCULATION OF THE OUTPUT LAYER.

N_s	P_1	P_2	P_3	P_4	P_5
$a_{2,1}^1$	1.158	0.6	0.92	-0.82	0.21
S_3	0.76	0.645	0.71	0.305	0.55

Finally, we use the sigmoid function to calculate the input of this last layer. We can see the decision results in the table 7. This table compares the results of our proposed prediction patients' model with the usual SMLP algorithm and shows the real decision from our ED database (EDDB) that was made for each patient P_i (1< i < 5).

TABLE VII. RESULTS COMPARISON

Methods	P_1	P_2	P_3	P_4	P_5
SMLP	0	0	0	0	1
DDMLP	1	1	1	0	1
EDDB	1	1	1	0	0

This simulation shows that our approach is effective on a small problem instance (5 patients): only one error compared to 4 for the standard model. This example details the calculation process within our proposed DDMLP algorithm in each step using D-S rules. In the following sub-section, we set up several experiments in order to compare our method with other classification techniques.

B. *Results and Discussion*

In this section, we perform two experimentations, the first one shows the contribution of our algorithm DDMLP compared to the SMLP and the second one compares our model to four well-known machine learning techniques. For the first experimentation, the parameter values used are: learning rate=0.001 and batch size= 5. Also, we run the BP algorithm of each method 20 epochs and report the average and standard deviation of classification accuracy in table 8. The results of this experiment show that the proposed model is more accurate compared to the SMLP and DMLP algorithm. We notice that our model obtained a high performance after 20 iterations but the SMLP algorithm is very far from the optimization.

TABLE VIII. RESULTS OF TESTS

Methods	Accuracy	Sensitivity	Specificity
SMLP	0.59	0.53	0.57
DMLP	0.71	0.69	0.67
DDMLP	0.83	0.79	0.81

In the second experimentation, we compare our DDMLP method to the Decision Tree (CART), Random Forest (RF), K-Nearest Neighbors (KNN) and LR algorithms (table 9). The ROC curve (Fig. 3) displays the results obtained from CART, KNN, RF, DDMLP and LR methods.

TABLE IX. RESULTS OF COMPARATIVE METHODS

Methods	Accuracy	Sensitivity	Specificity
CART	0.61	0.71	0.59
LR	0.74	0.70	0.75
RF	0.79	0.81	0.77
KNN	0.75	0.76	0.72
DDMLP	0.89	0.92	0.88

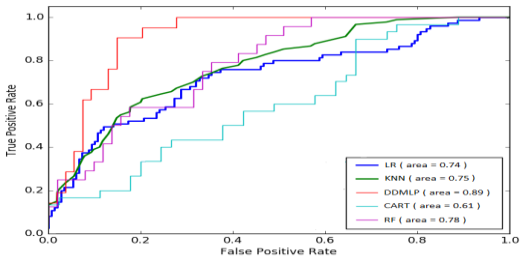


Fig. 3. Curve ROC of comparative methods.

Concerning the first experimentation, the table 8 shows that our method reduces errors. In fact, we obtain a more optimized network with fewer errors after 20 epochs (error correction) of the learning stage (BP algorithm). The SMLP and DMLP algorithms do not exceed 0.59 and 0.67 of accuracy and our proposed model gives an accuracy equal to 0.83. Indeed, the DMLP algorithm is better than the classical algorithm due to its dynamism and its management of missing data but the proposed model is more efficient than both. In addition, the table 9 and the ROC curve show the performance of our method according to these methods. Visually, our approach is more efficient and more accurate than other optimized methods. Also, the specificity and sensitivity of the other methods show that the rate of false positives and false negatives is very important

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compared to our method. This result can be argued by the management of uncertainty which improves the data quality of our model. This improvement can be explained by relieving the algorithm of missing data on one hand and by balancing the weight of the information according to its importance thanks to the D-S theory. So, the calculation of weights within out DDMLP algorithm helps to optimize the results (table 8) and minimizes the errors identified in the backward phase. Moreover, managing the uncertainty of used data improves the accuracy of our algorithm and reduces the number of false positives. Also, our study shows that machine learning can robustly predict hospital admission at ED triage. In addition, the data uncertainty measurement improves predictive reliability and performance significantly for used to inherent data source where highlight the need to incorporate the uncertainty into predictive models. Thus, the missing and unreliable data negatively influence the classification model which proves that our proposed approach handling the uncertainty is more efficient in reducing an errors rate than literature works. In addition, the learning speed and the complexity were among the major limits of the neuron networks but the dynamism and the new method of calculation of weight can improve them. However, this study presents some limits which can be improved in future studies, the use of textual data which can be more specific to the presentation of the patient state and for this, it is useful to add extraction technique to data textual in our proposed model. Second, getting the patient history is one of the most difficult tasks because each doctor completes it in a different way. So, a reliable, consistent and accurate database can better improve the performance and accuracy of classification results.

V. Conclusion

We propose in this paper a new DDMLP classification model based on DMLP and D-S theory. The comparison between our method outcomes over some known previous models shows the performances of our contribution. The improvement in results accuracy proves a powerful neuron network. Our proposed solution is considered as a decision support system for the benefit of the healthcare staff during triage step in ED in order to better manage resources and patients, and so reduce the ED overcrowding and improve patient care and so satisfaction. Moreover, this model is not exclusively limited to the health domain, it can be adapted to several classification problems which use uncertain data decreasing the error rate. However, it must be stressed that D-S theory is not effective in case of significant conflict between sources of information which brings us back to reviewing the relationships between the merged variables in order to manage this conflict. Thus, it can be appropriate to propose in future work an improvement of this theory within our proposed approach to take into account potential conflicts between different sources of information. In addition, we intend to generalize our algorithm in order to be able to vary the number of hidden layers and thus test the impact of this variation on the performance of our approach. The algorithm could then integrate this in its learning process and adapt the hidden layers according to the instance of the problem. Finally, as it was mentioned in the experimentation context section, other data are being recovered, that will allow us to enrich of experimentations in the context of our current national project.

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